Non-Functional Requirements for Restaurant Management

1.Performance Requirements

1. Response Time

- The system shall respond to user actions within 2 seconds for 95% of all requests.
- Critical operations (order placement, payment processing, table reservation confirmation) shall respond within 1 second.
- Dashboard loading time shall not exceed 3 seconds, even during peak hours.

2. Throughput

- The system shall support a minimum of:500 concurrent users per restaurant location.
    - 500 concurrent users per restaurant location.
    - 1,000+ orders per hour without performance degradation.
    - Kitchen Display System (KDS) shall process and display new orders in real time (<1 second delay).

3. Concurrent Users

- The system shall support simultaneous access by:
    - Customers (online ordering and reservations)
    - Staff (POS, KDS, inventory, staff management)
    - Management (reports and analytics)
- Performance shall remain stable with multiple user roles active at the same time.

4. Scalability

- The system shall scale horizontally to support:
    - Multiple restaurant branches
    - Increasing order volume during peak times (lunch/dinner hours).
- Adding new restaurants or users shall not require system downtime.

5. Availability & Uptime

- The system shall have a minimum availability of 99.9% uptime.
- Planned maintenance downtime shall be scheduled outside of peak business hours.
- The system shall automatically recover from failures without data loss.

6. Real-Time Processing

- Inventory updates shall reflect changes immediately after order completion.

- Order status updates (received, preparing, ready, delivered) shall be visible in real time to customers and staff.
- Loyalty points and rewards shall be applied instantly after order completion.

7. Data Processing & Reporting

- Standard reports shall be generated within 5 seconds.
- Large historical or analytics reports shall be generated within 15 seconds.
- Real-time sales and inventory dashboards shall update every 5–10 seconds.

8. Mobile Performance

- Mobile application performance shall be equivalent to the web application.
- Mobile screens shall load within 3 seconds on standard 4G/5G networks.
- The system shall function efficiently on low-bandwidth connections with graceful degradation**.**

9. Integration Performance

- POS, payment gateway, and third-party API integrations shall complete transactions within 2 seconds.
- API response times shall not exceed 1 second for standard requests.

10. Reliability Under Peak Load

- The system shall maintain full functionality during peak hours (weekends, promotions, holidays).
- No data inconsistency or transaction loss shall occur under high load conditions.

2.Security Requirements

1. Authentication

- The system shall require secure user authentication for all users.
- The system shall support strong password policies (minimum length, complexity, expiration).
- The system shall support multi-factor authentication (MFA) for administrators and sensitive roles.
- The system shall lock accounts after a configurable number of failed login attempts.

2. Authorization & Access Control

- The system shall implement role-based access control (RBAC).
- Users shall only access data and functions permitted by their assigned role.
- Administrative functions shall be restricted to authorized personnel only.
- The system shall prevent privilege escalation.

3. Data Confidentiality

- The system shall encrypt sensitive data at rest using industry-standard encryption (e.g., AES-256).
- The system shall encrypt data in transit using secure protocols (e.g., HTTPS/TLS).
- Sensitive information (passwords, payment data) shall never be stored in plain text.

4. Data Integrity

- The system shall ensure data is protected from unauthorized modification.
- The system shall validate all inputs to prevent data tampering.
- Audit mechanisms shall detect and log unauthorized data changes.

5. Audit Logging & Monitoring

- The system shall log all security-relevant events (logins, failed attempts, data changes).
- Audit logs shall be tamper-proof and accessible only to authorized users.
- The system shall support monitoring and alerting for suspicious activities.

6. Availability & Protection

- The system shall protect against common attacks such as:
    - SQL Injection
    - Cross-Site Scripting (XSS)
    - Cross-Site Request Forgery (CSRF)
- The system shall implement rate limiting to prevent brute-force attacks.
- The system shall ensure high availability and graceful recovery from failures.

7. Compliance & Privacy

- The system shall comply with applicable data protection regulations (e.g., GDPR, CCPA).
- User personal data shall be collected, stored, and processed according to privacy policies.
- The system shall support data deletion and anonymization upon request.

8. Backup & Recovery

- The system shall perform regular automated backups of critical data.
- Backups shall be encrypted and stored securely.
- The system shall support data recovery within a defined recovery time objective (RTO).

9. Session Management

- User sessions shall expire after a configurable period of inactivity.
- Session tokens shall be secure, unique, and protected from hijacking.
- Users shall be automatically logged out after session expiration.

3.Reliability Requirements

1. System Availability

- The system shall be available 99.9% of the time during business hours.
- The system shall remain operational during peak usage periods without failure.
- Planned maintenance shall not exceed predefined downtime limits.

2. Fault Tolerance

- The system shall continue operating in the event of partial component failure.
- Failure of one module (e.g., reporting) shall not affect core functions such as order processing.
- The system shall automatically switch to backup components when failures occur.

3. Error Handling

- The system shall handle unexpected errors gracefully without crashing.
- The system shall display user-friendly error messages without exposing technical details.
- Errors shall be logged for diagnostic and recovery purposes.

4. Data Consistency & Accuracy

- The system shall ensure data consistency across all modules (orders, inventory, payments).
- Transactions shall be atomic to prevent partial updates.
- In case of system failure, data shall not be lost or corrupted.

5. Recovery & Restart

- The system shall automatically recover from failures without manual intervention where possible.
- The system shall restart and restore services within a defined Recovery Time Objective (RTO).
- The system shall resume normal operations without data loss after recovery.

6. Transaction Reliability

- All transactions (orders, payments, reservations) shall be reliably completed or rolled back.
- Duplicate transactions shall be prevented during network or system interruptions.
- The system shall confirm successful completion of critical transactions.

7. Backup Reliability

- The system shall perform regular automated backups of all critical data.
- Backups shall be verified periodically to ensure successful restoration.
- The system shall allow data restoration to the most recent consistent state.

8. Monitoring & Alerts

- The system shall continuously monitor system health and performance.
- Automatic alerts shall be generated in case of failures or abnormal behavior.
- Monitoring data shall be retained for reliability analysis and improvement**.**

9. Scalability Under Load

- The system shall maintain reliable operation under increasing numbers of users.
- Performance degradation under high load shall not cause system failure.
- The system shall scale resources automatically to maintain reliability

4.Usability Requirements

1. Ease of Use

- The system shall be easy to use for users with minimal technical knowledge.
- New users shall be able to perform basic tasks (placing orders, reservations) without training.
- Common tasks shall require minimal steps to complete.

2. Learnability

- The system shall be learnable within a short time (e.g., less than 30 minutes for first-time users).

- The system shall provide clear labels, icons, and instructions.
- Tooltips and contextual help shall be available for complex features.

3. User Interface Consistency

- The system shall maintain a consistent layout, color scheme, and navigation across all screens.
- Similar actions shall follow the same interaction patterns throughout the system.
- Standard UI conventions shall be followed.

4. Accessibility

- The system shall be accessible to users with disabilities.
- The system shall support keyboard navigation and screen readers.
- Text size and contrast shall be adjustable for readability.

5. Responsiveness

- The system shall be responsive across desktops, tablets, and mobile devices.
- User interactions shall provide immediate feedback (loading indicators, confirmations).
- Screen transitions shall be smooth and intuitive.

6. Error Prevention & Recovery

- The system shall minimize user errors through input validation and clear guidance.
- Clear and helpful error messages shall be displayed when errors occur.
- Users shall be able to easily correct mistakes without losing data.

7. User Satisfaction

- The system shall provide a visually appealing and clean interface.
- The system shall minimize user frustration and cognitive load.
- User satisfaction levels shall meet defined usability benchmarks.

8. Help & Support

- The system shall provide in-app help, FAQs, and user guides.
- Help content shall be easy to search and understand.
- Support options shall be clearly visible within the interface.

9. Personalization

- The system shall provide role-based dashboards tailored to different users.
- Users shall be able to customize basic preferences (language, notifications).
- Frequently used features shall be easily accessible.

5.Maintainability Requirements

- **Modular Design**
  The system shall be designed using a modular architecture so that individual components can be modified, replaced, or enhanced without impacting other parts of the system.
- **Code Readability & Standards**
  The source code shall follow established coding standards and naming conventions to ensure readability and ease of understanding for future developers.
- **Documentation**
  The system shall include up-to-date technical documentation, including system architecture, API documentation, and inline code comments to support maintenance activities.
- **Ease of Bug Fixing**
  The system shall allow defects to be identified, isolated, and fixed with minimal impact on system functionality and availability.
- **Configurability**
  System configurations (e.g., business rules, thresholds, UI labels) shall be configurable without requiring code changes or system redeployment.
- **Logging & Error Handling**
  The system shall provide comprehensive logging and error-handling mechanisms to support troubleshooting and maintenance.
- **Testability**
  The system shall support automated unit, integration, and regression testing to ensure changes can be validated efficiently.
- **Update & Upgrade Support**
  The system shall support regular updates and enhancements without requiring complete system downtime.
- **Dependency Management**
  External libraries, APIs, and third-party dependencies shall be well-documented and version-controlled to simplify maintenance and upgrade.
- **Maintainability Metrics**
  The system shall maintain a low complexity level (e.g., cyclomatic complexity within acceptable limits) to ensure long-term maintainability.

6. Environmental and Operational Requirements

1.Environmental Requirements

These are the conditions in which the system must work.

- The system should operate reliably in restaurant environments with noise, heat, and frequent human interaction.
- Hardware used (POS terminals, tablets, printers) should work in temperature ranges typically found in kitchens and dining areas.

- System components should tolerate power fluctuations and have protection against sudden shutdowns.
- RMS should support stable internet connectivity but also provide limited offline support during temporary network failures.
- Devices should function properly under continuous usage for long working hours.
- The system should be compatible with commonly used operating systems (Windows, Linux, Android, iOS).
- Network components should operate properly in environments with multiple Wi-Fi devices and interference.
- Environmental factors such as dust, humidity, or minor spills should not affect normal device operation.
- Screens and terminals should have sufficient brightness and visibility under different lighting conditions.

2.Operational Requirements

These describe how the system should be used and managed during daily operations.

- The system should support multi-user operations (admin, manager, cashier, waiter, chef).
- The system must run 24/7 during restaurant working hours without failure.
- Login should be role-based with secure authentication and access control.
- System should provide automatic data backup at scheduled intervals.
- Regular software updates and maintenance must be supported without disrupting operations.
- RMS should integrate smoothly with POS machines, printers, barcode scanners, KDS, and payment gateways.
- System must process orders, billing, and inventory updates in real time with minimal delay.
- The interface must be simple, fast, and easy to use by non-technical staff.
- Error messages should be clear and user-friendly, not technical.
- System should support training mode or demo environment for new staff.
- Audit logs should record user activities, transactions, and configuration changes.
- The system should allow daily report generation for sales, inventory, and staff operations.